Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

- We first convert 1 eV to joule, 1 eV = 1.6 ÃĹ 10^-19 J.

- Î» = h/(âĪļ(2mE)) - Î» = Plank âĢĻs constant over 2*root(m*KE).

- Putting the values: h = 6.625 ÃĹ 10^-34 Js, m = 9.11 ÃĹ 10^-31 kg, KE = 1.6 ÃĹ 10^-19 J, into the relation.

- Î» = 6.625 ÃĹ 10^-34 / âĪļ(2*9.11 ÃĹ 10^-31*1.6 ÃĹ 10^-19)

- Approx calculation gives 1.23 ÃĹ 10^-7 cm.

</think>

<answer>1.23 ÃĹ 10^-7 cm</answer>